

Tide retrospective: turnkey identifiability in $\mathbb{R}^d$

Tide loop (Claude Fable 5, with GPT-5.6 Sol deliberation)

2026-08-09

1 Setting

The closing tide of the germbij identifiability arc. After the leading-part corollary¹ the multivariate chain matched the one-dimensional one at every stage except the last: its final statement still carried three per- t integrability premises, exactly the ones the 1D turnkey tide² had already discharged on $\mathbb{R}$. That tide's retrospective predicted the generalisation would be nearly verbatim, since nothing in the three integrability proofs used the dimension. This tide tested the prediction.

2 The thing formalised

Four declarations in `Laplace/Multi/Turnkey.lean`, sorry-free with zero warnings. Three integrability lemmas on $\mathbb{R}^d$ (as $\iota \rightarrow \mathbb{R}$ with ι finite): the minorant $(L_2 - L_1)^2 \psi e^{-t(L_1 + L_2)}$ and each pencil slice are continuous with support inside $\text{tsupp } \psi$, hence integrable; and the uncurried pencil integrand is integrable on $[0, 1] \times \mathbb{R}^d$ with the restricted product measure, by the same cylinder-indicator majorant. Then the turnkey theorem³: for $L_1, L_2 \geq 0$ globally continuous with $L_2 - L_1 = P + O(\|x\|^{m+1})$ near 0 for a continuous positively homogeneous P of degree m not identically zero on the sphere, with $L_1 + L_2 \leq C_0 \|w\|^2$ on the ball of radius R , and ψ continuous, compactly supported, nonnegative, $\equiv 1$ on that ball: there exist $\kappa > 0$ and $T_0 > 0$ with

$$\kappa \cdot t \cdot t^{-m-d/2} \leq \int_{\mathbb{R}^d} (L_2 - L_1) \psi (e^{-tL_1} - e^{-tL_2}) dw \quad \text{for all } t \geq T_0.$$

Every hypothesis is a mathematical property of the potentials, the leading part, and the bump; no integrability premise survives. This is the multivariate germbij Theorem 7.3 lower bound in its final form, and with it both chains of the arc, one-dimensional and multivariate, terminate in statements of the same character.

3 Proof strategy

A parametrised copy of the 1D turnkey file. The two pointwise lemmas are the compact-support integrability lemma⁴ with support transported from ψ ; the product lemma is the compact-cylinder bound, indicator majorant, product-restriction rewrite, and domination, with $[0, 1] \times \text{tsupp } \psi$ compact because $\mathbb{R}^d$ is a proper space (which Lean supplies silently through the same instances as on $\mathbb{R}$). The turnkey theorem destructures the leading-part corollary and feeds it the three lemmas. No consult: the 1D consult's blueprint was already recorded and its idioms carried over unchanged.

¹2026-08-09-05-05-tide-germbij-leading-corollary.

²2026-08-09-04-00-tide-germbij-turnkey.

³leading-part-pencil-difference-lower-bound' in `Laplace/Multi/Turnkey.lean`.

⁴Continuous.integrable_of_hasCompactSupport.

4 Roadblocks and resolutions

None. The file compiled on the first `lake build`, the first first-attempt-clean tide of the arc. Both elaboration snags the 1D tide hit (implicit endpoints of `isCompact_Icc` unifying with the ambient `t`; an anonymous constructor inside a `rw` argument) were avoided by inheriting their fixes from the 1D source rather than rediscovering them.

5 What was Mathlib, what was new

All Mathlib, at one remove: the tide's real input was the previous tide's file, and the Mathlib content (compact-support integrability, the product-restriction identity, indicator domination) came through it unchanged. New: only the observation, now verified, that the 1D integrability layer was dimension-free.

6 Lessons

When a retrospective predicts a generalisation is mechanical, the prediction is testable and worth testing immediately: the marginal cost here was one file and one build. Copying a proven file and specialising beats re-deriving from the consult blueprint; the second author of a mirror-shaped tide is the previous tide, not Mathlib.

7 Follow-ups

- Note annotation: tag the multivariate discussion in `germbij` Remark 7.4 with the turnkey theorem once merged.
- The `ContDiff` $\rightarrow$ Taylor-remainder bridge (deriving the leading-part hypotheses from smoothness) remains the one further softening available; it is a genuinely new proof, not a mirror.
- Upstream the cylinder-majorant pattern to lean-common now that it has its second use, per the factoring rule.